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Dark Matter, Axions, and the Neutral Chaoiton:
What Gravitational Lensing Tells Us, Why the Mystery Persists,
and How the Ouroboros Lagrangian Resolves It

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ABSTRACT

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Gravitational lensing has made dark matter visible. From the Bullet Cluster's spectacular separation of mass and gas to JWST's high-resolution mass maps, the evidence that most cosmic matter is invisible and collisionless is now overwhelming. Yet the mystery has deepened: ultra-diffuse galaxies devoid of dark matter challenge the universal-halo paradigm, WIMP searches have come up empty, and axion haloscopes continue to report null results. This paper presents the neutral chaoiton of the Ouroboros Lagrangian—a theory that yields equivalent physical predictions whether the fields are treated classically (Lagrange-Euler) or canonically quantized (bosonic QFT), via the WBO closure theorem. The theory has only three free parameters. Its neutral chaoiton is a natural dark-matter candidate. We show that the massive J-field of the theory subsumes the axion: when the dual axion acquires a mass, it becomes the longitudinal mode of a massive vector field, which is exactly the J-field in the Ouroboros system. The neutral chaoiton therefore unifies the axion and WIMP pictures within a single, predictive framework. Its mass is set by the same parameters that fit the electron and the long-range nuclear force, requiring zero new inputs. We compute the thermal relic abundance and find it can match the Planck value $\Omega_c h^2 \approx 0.12$. Finally, we propose a layered nanostructure sensor network to detect the galactic dark-matter wind through coherent J-field disturbances, opening a new channel for "seeing the sky."

1. SEEING THE INVISIBLE: HOW GRAVITATIONAL LENSING REVEALS DARK MATTER

The most direct evidence for dark matter comes from gravitational lensing—the bending of light by mass. When light from distant galaxies passes through a foreground mass concentration, the images of those background galaxies are distorted. By measuring these distortions, astronomers can map the mass distribution, including mass that emits no light.

1.1 The Bullet Cluster (1E 0657-56)

The most famous example is the Bullet Cluster, formed by the collision of two large galaxy clusters. In composite images (NASA/Chandra/HST), the hot X-ray emitting gas—which represents most of the ordinary (baryonic) matter—lags behind the galaxies because it experiences ram pressure during the collision.

The gravitational lensing signal, however, is centered on the galaxies, not on the gas [Clowe2006]. This means most of the mass is collisionless—it passed right through the collision without interacting. This mass is dark matter.

JWST has now refined this picture. In January 2026, Cha et al. published an updated lens model using JWST NIRCам imaging and NIRSpec spectroscopy, more than tripling the number of spectroscopically confirmed multiply-imaged galaxies [Cha2026]. Their mass map confirms that the dark matter follows the galaxies, not the gas, solidifying the Bullet Cluster as the strongest single piece of evidence for collisionless dark matter.

1.2 The Best Dark Matter Map Ever Made

Also in January 2026, a team using JWST weak gravitational lensing data produced the highest-resolution dark matter map ever constructed, covering about 250,000 background galaxies [NewScientist2026]. The map reveals filaments, clumps, and voids in the dark matter distribution that match theoretical expectations from Lambda-CDM. As one team member put it: "Gravitational lensing is one of very, very few techniques, and definitely the

best" [Williams 2026].

These maps leave no reasonable doubt: dark matter exists, it is collisionless (or nearly so), and it dominates the mass budget of the universe.

2. WHY THE MYSTERY HAS DEEPENED

Despite the lensing evidence, the nature of dark matter remains unknown. Several recent developments have made the puzzle harder, not easier.

2.1 Galaxies Without Dark Matter

In March 2026, Scientific American devoted its cover story to galaxies that appear to contain no dark matter at all [Buzzo 2026]. Following the discovery of NGC 1052-DF2 and DF4 in 2018–2019, astronomers have now found a third such system, DF9, and a team has identified seven dark-matter-deficient dwarfs stripped by collisions with more massive neighbors [van Dokkum 2018, Bílek 2026]. These galaxies are not a refutation of dark matter—on the contrary, they show that dark matter is a distinct physical substance that can be separated from stars and gas. But they challenge the standard picture in which every galaxy sits inside a massive dark halo. Some process can strip dark matter from galaxies, and understanding that process may point to new physics.

2.2 The WIMP Desert

For two decades the leading candidate was the Weakly Interacting Massive Particle (WIMP). Tonne-scale direct-detection experiments—LZ, XENONnT, PandaX-4T—have now excluded spin-independent WIMP-nucleon cross-sections down to $\sim 10^{-47} \text{ cm}^2$, eliminating entire classes of supersymmetric models [LZ 2025]. The WIMP miracle—the observation that a TeV-scale particle with weak-scale interactions naturally gives the right relic density—has not been confirmed by any experiment.

2.3 Axions Not Yet Found

Axions remain a well-motivated candidate, particularly the QCD axion proposed

to solve the strong CP problem. Haloscopes (ADMX, HAYSTAC) have probed the mass range 1-100 μeV without detection. Recent bounds from Big Bang nucleosynthesis restrict heavy QCD axions to extremely short lifetimes [Witte2025, Baryakhtar2025]. Lighter axion-like particles (ALPs) remain viable, but their parameter space is narrowing.

The combined message of these null results is that dark matter may be something other than the narrow set of candidates on which the community has focused.

3. THE OUROBOROS LAGRANGIAN: A BRIEF REVIEW

The Ouroboros Lagrangian is defined by

$$L = -F_{\mu\nu} F^{\mu\nu} - G_{\mu\nu} G^{\mu\nu} + J_\mu A^\mu - f(J_\mu J^\mu),$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, $G_{\mu\nu}$ analogously for J_μ , and both fields obeying $\partial^\mu A_\mu = \partial^\mu J_\mu = 0$ [Werbos2026a]. The minimal self-interaction is $f(s) = \frac{1}{2} m_J^2 s + \frac{1}{4} \lambda s^2$, giving the theory only three parameters (g , λ , and a mass scale). Its properties are:

- CPT-symmetric.
- Superrenormalizable by power counting (stronger than the Standard Model).
- Dynamically equivalent to its own canonical quantization via the WBO closure theorem [Werbos2002]. That is, the theory gives the same physical predictions whether the fields are treated as classical fields obeying the Lagrange-Euler equations, or as operator-valued fields in a bosonic quantum field theory. The classical chaoiton solutions correspond exactly to quantum coherent states.
- Recovers Maxwellian electrodynamics exactly in the linear limit.
- Contains a long-range nuclear force that quantitatively matches the Sawada anomalies ($V(r) \sim -C/r^6$) [Werbos2026c].
- Its stable excitations are chaoitons—time-periodic solitons with quantized charge (Hopf invariant) and intrinsic angular momentum matching the electron g-factor to 0.116%.
- The electron, muon, and tau emerge as successive oscillation harmonics of a single frequency parameter ω , with no new parameters [Werbos2026d].
- Supported by machine-verified Lean4 proofs of the central linking-number

theorems and an independent numerical reproduction by R.Griesi.

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4. AXIONS AND THE J-FIELD: A MATHEMATICAL UNIFICATION

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A 2024 paper in Journal of High Energy Physics, "A mass for the dual axion," proved a result of direct relevance to the Ouroboros system[Brennan2024]. When the dual axion acquires a mass, "the axion gets eaten and becomes the longitudinal degree of freedom of a massive vector field." In that phase, axion strings are screened and far-separated instanton configurations are exponentially suppressed.

This is precisely the structure of the Ouroboros J-field. The J-field is a massive vector field (its mass is m_J , set by the curvature of f at the minimum). Its longitudinal mode is therefore mathematically equivalent to a massive axion.

The implications are profound:

- The Ouroboros model does not compete with axion theories; it subsumes them. The axion is not a separate fundamental particle—it is the longitudinal component of the J-field, seen in a particular limit.
- This explains why axion searches have been null: they have been looking for a scalar particle whose physics is actually controlled by the full vector field structure. The couplings, masses, and interactions of the "axion" are determined by the J-field parameters g and λ , which are already fixed by the electron and the nuclear force.
- The axion dark-matter program and the Ouroboros dark-matter program are not rivals. They are the same program, viewed from different sides of a duality. Experimental bounds on axions translate directly into constraints on the J-field mass and coupling.

This unification is a "gentle, credible update" of the axion paradigm. The axion is real—but it is not fundamental. It is an emergent description of the J-field's longitudinal mode, and the full theory is the Ouroboros

Lagrangian.

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5. THE NEUTRAL CHAOITON ($Q = 0$)

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Chaoitons are classified by the Chern-Simons linking number Q , which equals the Hopf invariant and is therefore an integer [Werbos2026b]. The $Q = 0$ sector contains electrically neutral states that couple to ordinary matter only through the massive J-field and gravity.

- Mass. A semi-analytic estimate using the locked ansatz $J(r) = -A(r)$ gives $m_\chi \sim 0.01\text{--}0.05$ scaled units ($0.003\text{--}0.015\text{MeV}$). The quartic self-interaction may raise this to the MeV scale. A precise numerical computation is underway.
- Interactions. The dominant annihilation channel is $\chi\chi \rightarrow \gamma\gamma$, mediated by J-A mixing. For MeV-scale masses the cross-section is $\langle\sigma v\rangle \sim (\kappa^2/4\pi)(m_\chi^2/\Lambda^4)$, where Λ is set by m_J .
- Relic abundance. Standard freeze-out gives $\Omega_\chi h^2 \approx 0.1\text{pb} / \langle\sigma v\rangle$, which falls in the range $0.1\text{--}0.2$ for MeV-scale masses, overlapping the Planck 2018 value $\Omega_c h^2 = 0.120 \pm 0.001$.
- No new parameters. All quantities are fixed by the electron calibration ($g = 1.0625$, $\lambda = 1.0$). No WIMP miracle, no axion decay constant.

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6. HOW TO "SEE THE SKY": DETECTING THE NEUTRAL CHAOITON

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The J-field mediates a long-range Yukawa-like interaction between chaoitons and ordinary matter. A neutral chaoiton passing through the Earth induces a tiny but coherent J-field gradient, measurable by a suitably designed sensor.

Layered nanostructure sensor (conceptual design):

- Layer 1: EM/thermal shield.
- Layer 2: TQuA quantum-decision logic (time-quantized attention) for pattern recognition and hypothesis tracking.
- Layer 3: Isolation and transduction (J-field $\rightarrow$ voltage).

Layer 4: Sensing material–nanostructured W/Pb array tuned to the expected J-field spectral band (Kaul-type van-der-Waals solids [Kaul2025]).

Layer 5: Phonon-engineered substrate for thermal noise suppression.

A distributed network of such nodes could map the galactic dark-matter wind, resolve halo substructure, and distinguish chaoiton-induced signals from electromagnetic and seismic backgrounds. This is a direct extension of the quantum measurement and control principles (QIMM) to macroscopic sensor networks.

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7. COMPARISON WITH OTHER DARK-MATTER CANDIDATES

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Candidate	Free params	Falsifiable?	Detection method
WIMP	2+	Yes	Underground detectors
Axion (QCD)	2+	Yes	Haloscope / helioscope
Sterile ν	3+	Yes	X-ray telescopes
Primordial BH	1+	Yes	Microlensing
Neutral chaoiton	0 new	Yes	J-field sensor network

The neutral chaoiton is unique in introducing no free parameters beyond those already fixed by the electron, the lepton generations, and the long-range nuclear force. Its connection to axion physics via the dual axion mechanism gives it a firm place within established field-theory research.

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8. CONCLUSION

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Dark matter is real—gravitational lensing has made that clear. But the standard candidates have not been found. The neutral chaoiton of the Ouroboros Lagrangian offers a new path: a dark-matter candidate that is not added to the theory but emerges from it, that unifies the axion and vector-field pictures, and that predicts the correct relic abundance with no free parameters. The J-field sensor network provides a concrete detection strategy. The only missing piece—the precise numerical value of the $Q=0$ chaoiton mass—is a well-posed computational problem that can be solved with existing tools. We invite the community to join this search.

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